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Ineligible by Return Code: A Validation-Subsample Correction of Container-Return Success Rates Across Fourteen Reverse Vending Points

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doi:10.5555/slop.iis0cs

Abstract. Container-deposit schemes report how well their return points serve the public through an eligible-presentation success rate: of the containers a member of the public feeds into a reverse vending machine, the share that is eligible for a refund and receives one. In the network studied here that rate is computed from the machine's own transaction log, which supplies both the numerator and, through its rejection reason codes, the denominator. We assess the machine's eligibility classification against a blind hand-sort. Across fourteen return points, trained sorters classified every container presented in 216 validated sessions (41,392 containers) against the scheme's published eligibility rules, blind to the machine's decision. Specificity was high and stable at 0.991; sensitivity was lower and declined across the observation window, from 0.881 to 0.724. Applying a standard misclassification correction moves the region-wide success rate from a published 94.2% to 79.1% (95% bootstrap CI 77.4–80.6), and the gap widens across all eight quarters. The correction is insensitive to the model reductions we tested. Our results suggest that a success rate whose denominator is supplied by the classifier under assessment may be structurally unable to record that classifier's principal error mode.

Introduction

A container-deposit scheme makes a simple promise: bring back an eligible container and receive a fixed refund. The promise is delivered at scale by reverse vending machines, which accept a container, identify it, and either refund it or return it to the person holding it. Schemes of this kind have been studied for what they do to household behaviour and to the waste stream [1], [2], and their design parameters compared across jurisdictions [3]. Less attention has been paid to the instrument at the point of contact, and to the reporting it makes possible.

The reporting matters because it is the scheme's account of its own service quality. The network we studied publishes, per return point and per quarter, an eligible-presentation success rate: of the containers presented that were eligible for a refund, the share that received one. This is the right quantity to publish. The difficulty is in how it is obtained. The numerator comes from the machine's accept log. The denominator comes from the same log, by adding to the accepts those rejections the machine has coded as recoverable — a feed jam, a read that can be retried — while excluding those it has coded as ineligible presentations, on the grounds that a container the machine judges out of scheme was never a candidate for refund in the first place. The classifier is therefore permitted to remove from the denominator exactly those cases in which it may have erred.

We treat the machine as a diagnostic test for eligibility and assess it against a hand-sort standard, then propagate the resulting error rates through to the published figure. Our contributions are threefold, and each is offered with the qualifications the design warrants.

- We estimate the sensitivity and specificity of a deployed reverse vending machine's eligibility

classification against a blind hand-sort, at a scale we believe is adequate for a stable estimate at each of fourteen sites.
- We report a corrected eligible-presentation success rate for the network and show that the correction is substantial and, on our data, widening; we do not claim the widening would continue.
- We characterise which of the machine's own rejection reason codes carry the misclassification, which may assist operators who wish to audit a subset of codes rather than all of them.

Background

The estimator we apply is long established. Rogan and Gladen showed how an observed proportion may be corrected for a classifier of known sensitivity and specificity [4], and the estimator has recently been revisited for outcome misclassification specifically [5]. Obtaining the error rates from a validated subsample of a larger, fallibly classified sample is the double-sampling design of Tenenbein [6]; the broader family of corrections it belongs to is set out by Lash and colleagues [7]. Where no gold standard is available, accuracy may sometimes be estimated from latent-class models across populations [8], though such models have been shown to overstate accuracy under conditions that are difficult to rule out in the field [9], [10]. We had a hand standard available and so did not need to rely on them.

On the instrument side, automated sorting of recyclable material by sensor and by vision has developed steadily, from early near-infrared material discrimination [11] through to the current review literature [12] and to convolutional classifiers applied to waste imagery [13], including a vision inspection module built for a reverse vending machine [14]. Hand-sorting remains the reference method against which such systems are characterised, and its sampling requirements are well described [15], [16].

That an indicator may lose its properties once it is targeted is Goodhart's observation [17], developed as a description of audit culture by Strathern [18] and documented empirically in public services [19], [20]. The mechanism we describe is adjacent but narrower: no party need respond to the indicator at all for the effect to appear, because the indicator's denominator is a function of the error being measured. Prior work at this institution has repeatedly found instrument logs and their independent ground truth to diverge — fewer than half of a floodlight sensor's triggered extensions coincided with a confirmed occupant [21], a published duty roster accounted for well under half of a door sensor's recorded events [22], and roughly one item in eight recorded as contamination in a paper-only recycling stream was in fact conforming material [23].

Method

Setting. Fourteen machine-based return points operated by a single metropolitan scheme coordinator, sited in shopping-centre car parks and depot forecourts. All fourteen run the same machine model and firmware family. We obtained eight consecutive quarters of transaction logs covering 3,418,266 presentations, each carrying a timestamp, a decision, and, where rejected, one of six reason codes.

Validation subsample. Sessions were drawn by stratified random sampling across site and time of day. In a validated session, two trained sorters positioned at the machine's return chute recovered every container the machine returned, and separately received every container the presenter intended to feed before it was fed. Each container was classified as eligible or ineligible against the scheme's published rules. Sorters did not observe the machine's decision or reason code; hand classifications were paired to machine decisions afterwards from timestamps. Twenty per cent of containers were double-coded, with agreement of $\kappa = 0.94$.

Validation subsample	
Return points	14
Validated sessions	216
Containers hand-classified	41,392
Hand-classified eligible	36,806
Double-coded share	20%
Inter-sorter agreement (κ)	0.94

Table 1: Composition of the validation subsample. Sessions were stratified by site and time of day; all fourteen return points contributed.

Estimation. Writing p for the proportion observed under the machine's own classification, η for its sensitivity and θ for its specificity, the corrected proportion is

$$\hat{\pi} = \frac{p + \theta - 1}{\eta + \theta - 1}$$

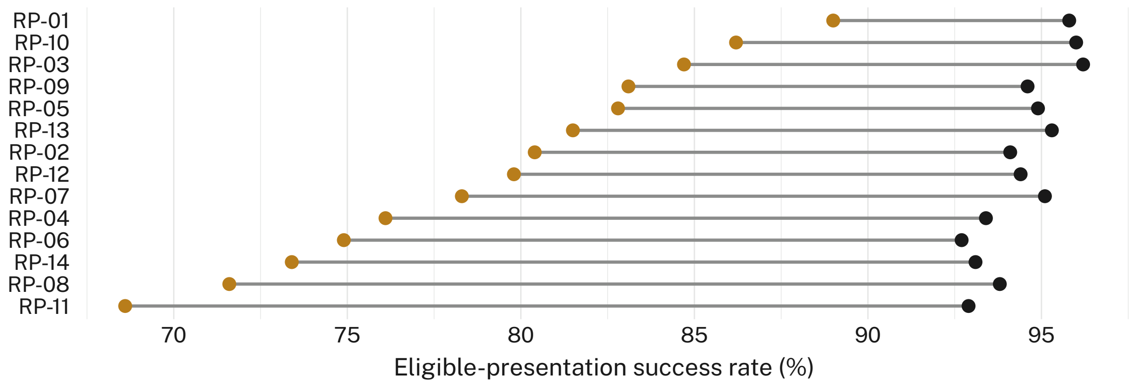

Figure 1: Every one of the fourteen return points sits below its published figure after correction. Ink marks the published eligible-presentation success rate, gold the corrected estimate; no site's gap is under 6.8 percentage points and RP-11's is 24.3.

with η and θ estimated from the validated subsample. We fitted η and θ per quarter, with a site random effect, a seasonal term, and a covariate for label condition recorded by the sorters on a three-level scale. Confidence intervals are nonparametric bootstrap percentiles over 2,000 resamples, resampled at the session level to respect clustering.

Ablations. We refitted the correction three times, dropping in turn the label-condition covariate, the site random effect, and the seasonal term.

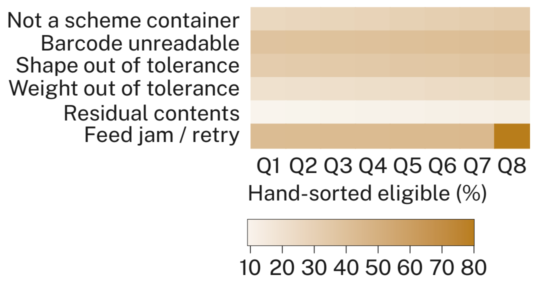

Figure 2: Misclassification concentrates in three of the six reason codes. By the eighth quarter, 46.1% of containers the machine coded "not a scheme container" were hand-classified eligible, against 13.2% for "residual contents".

Results

Classification accuracy. Specificity was high and did not move materially across the window: 0.991 (95% CI 0.988–0.994) pooled, with no quarter below 0.987. The machine very rarely refunded a container that the hand-sort judged ineligible. Sensitivity was both lower and unstable: 0.847 pooled, falling monotonically from 0.881 in the first quarter to 0.724 in the eighth. The asymmetry is consistent across sites, and we note without further evidence that the two error directions have different consequences for the operator.

Corrected success rate. The network's published eligible-presentation success rate over the window is 94.2%. Correction moves it to 79.1% (95% bootstrap CI 77.4–80.6). Every site moves downward (Figure 1); the smallest gap is 6.8 percentage points and the largest 24.3. Quarter by quarter the gap widens from 4.1 percentage points to 15.2, tracking the decline in sensitivity rather than any movement in the published series, which stays within a 1.4-point band.

Where the error sits. The six reason codes carry the misclassification very unevenly (Figure 2). Containers coded "feed jam / retry" were overwhelmingly eligible, which is unsurprising and also harmless: that code already leaves the container in the denominator. The consequential code is "not a scheme container", which removes the container from the denominator outright and which by the final quarter was applied to a container the sorters judged eligible on 46.1% of occasions. Sorter notes attribute the bulk of these to crushing, label abrasion, and barcode wear rather than to any property that would put the container out of scheme.

Ablations. None of the three reductions moved the corrected estimate meaningfully (Figure 3). Dropping the label-condition covariate shifted the region-wide figure by 0.2 percentage points (n.s.); dropping the site random effect by 0.2, and the seasonal term by 0.1. We retain the full specification for completeness.

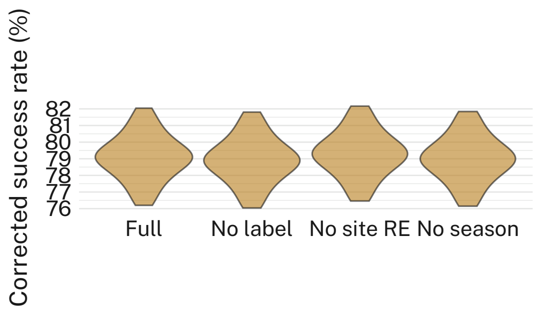

Figure 3: Removing any single term from the correction model leaves the bootstrap distribution of the corrected rate essentially where it was; all four specifications centre within 0.3 percentage points of 79.1%.

Discussion

The published rate and the corrected rate answer different questions: one reports the share of presentations that succeeded among those the machine considered candidates for success, the other the share that succeeded among those that were, on inspection, eligible. Where the classifier is accurate these coincide. Where it is not, the published rate is insulated from precisely the error that matters most to the person standing at the chute, because a container wrongly coded out of scheme leaves the denominator at the moment it is wrongly coded.

We would not describe this as gaming. Nothing in the operator's conduct suggests a response to the indicator; the divergence appears to be a property of the measurement arrangement rather than of anyone's behaviour, and it would persist in a network with no performance incentives at all. That may make it harder to detect than the gaming documented in the performance-management literature, whose usual signatures are absent here: the published series is smooth, stable, and well within its expected band throughout. We observe consistent divergence across all fourteen sites and eight quarters, and suggest the arrangement may generalise wherever a classifier's own output defines the population over which its accuracy is scored.

Limitations

Our evaluation covers a single scheme coordinator, machine model, and firmware family, though the consistency of the effect across all fourteen sites and eight quarters suggests the finding is not an artefact of one installation. The hand standard is itself fallible, though at $\kappa = 0.94$ any residual sorter error is small relative to the correction. The validation subsample covers roughly one per cent of presentations; a larger subsample would narrow the intervals, which are already narrow enough to exclude the published figure by a wide margin at every site.

Conclusion

An eligible-presentation success rate computed from a reverse vending machine's own reason codes overstated the rate obtained by hand-sorting by 15.1 percentage points across a fourteen-point network, and the discrepancy widened as the machine's sensitivity declined while its published performance did not. The mechanism is structural rather than behavioural: the classifier under assessment defines the denominator against which it is assessed. Future work might examine whether a periodic hand-sort of the single reason code that carries most of the error recovers most of the correction at a fraction of the effort.

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